

Tomohiko Nakamura

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Research Interests

Signal-processing-inspired deep learning, audio and music signal processing, and machine learning

Job

Senior Researcher <i>The National Institute of Advanced Industrial Science and Technology (AIST), Japan.</i>	Apr. 2023–Present
Project Research Associate <i>Graduate School of Information Science and Technology, The University of Tokyo, Japan.</i>	Sept. 2019–Mar. 2023
Researcher <i>Intelligent Systems Laboratory, SECOM, Japan.</i>	Apr. 2016–Aug. 2019
Research Fellow (DC2) <i>Japan Society for the Promotion of Science (JSPS), Japan.</i>	Apr. 2015–Mar. 2016

Education

Ph.D. degree in Information Science and Technology <i>Graduate School of Information Science and Technology, The University of Tokyo, Japan.</i>	Mar. 2016
Master's degree in Information Science and Technology <i>Graduate School of Information Science and Technology, The University of Tokyo, Japan.</i>	Mar. 2013
Bachelor's degree in Engineering <i>Faculty of Engineering, The University of Tokyo, Japan.</i>	Mar. 2011

Teaching

Applied Gaussian Process and Machine Learning <i>Graduate School of Information Science and Technology, The University of Tokyo, Japan.</i>	6, Dec. 2021
Advanced Signal Processing <i>Graduate School of Information Science and Technology, The University of Tokyo, Japan.</i>	23, June 2020 and 21, June 2022, June 2024
Student Experiment <i>Department of Mathematical engineering and information physics, The University of Tokyo, Japan.</i>	Apr. 2020–Mar. 2023

Languages

Japanese (native), English (basic)

Competitive Funds

Variable time-resolution audio processing framework adaptive to acoustic scenes <i>JSPS PRESTO (Fundamental Innovation for Real-World Intelligent Systems)</i>	Apr. 2023–Mar. 2027
Development of deep-layered analysis-by-synthesis techniques for acoustic scene analysis with human intervention <i>JSPS KAKENHI</i>	Apr. 2023–Mar. 2027
Sampling-frequency-independent deep learning for audio media processing <i>JST ACT-X (Frontier of Mathematics and Information Science)</i>	Oct. 2021–Mar. 2024
+ 4 funds as representative, 5 funds as co-researcher, and 1 fund as research participants.	

Publications

Journal Papers.....

- [1] Ren Uchida, Kohei Yatabe, and Tomohiko Nakamura, "Encoder-masking-decoder networks using orthogonal convolutional layer as invertible linear encoder," *Acoustical Science and Technology*, vol. advpub, no. e26.10, May 2026.
- [2] Kanami Imamura, Tomohiko Nakamura, Norihiro Takamune, Kohei Yatabe, and Hiroshi Saruwatari, "Stride conversion algorithms for convolutional layers and its application to sampling-frequency-independent deep neural networks," *Signal Processing*, vol. 242, Nov. 2025.
- [3] Yuki Ito, Tomohiko Nakamura, Shoichi Koyama, Shuichi Sakamoto, and Hiroshi Saruwatari, "Spatial upsampling of head-related transfer function using neural network conditioned on source position and frequency," *IEEE Open Journal of Signal Processing*, vol. 6, pp. 1109–1123, Sept. 2025.
- [4] Yusaku Mizobuchi, Daichi Kitamura, Tomohiko Nakamura, Norihiro Takamune, Hiroshi Saruwatari, Yu Takahashi, and Kazunobu Kondo, "Music bleeding-sound reduction based on time-channel nonnegative matrix factorization," *APSIPA Transactions on Signal and Information Processing*, vol. 14, no. 1, e18, July 2025.
- [5] +10 papers

Peer-Reviewed International Conferences and Workshops.....

- [1] Woan-Shiuan Chien, Tomohiko Nakamura, Huan-Yu Chen, Fukayama Satoru, Hitoshi Suda, Jun Ogata, and Chi-Chun Lee, "Two-sided fairness transfer for gender-neutral speech emotion recognition with partially observed attributes," in *Proceedings of INTERSPEECH*, Sept. 2026.
- [2] Daigo Takizawa, Tomohiko Nakamura, Samuele Cornell, William Chen, Satoru Fukayama, and Shinji Watanabe, "Dissecting sensitivity to training language in self-supervised speech learning using neural audio codec tokens," in *Proceedings of INTERSPEECH*, Sept. 2026.
- [3] Shinnosuke Takamichi, Tomohiko Nakamura, Hitoshi Suda, Satoru Fukayama, and Jun Ogata, "MangaVox: Dataset of acted voices aligned with manga images towards computer understanding of audio comics," in *Proceedings of IEEE International Conference on Acoustics, Speech, and Signal Processing*, May 2026, pp. 19467–19471.
- [4] +42 papers

Invited Talks.....

- [1] Tomohiko Nakamura, "Trends and prospects for audio source separation using deep learning," *Meeting on Technical Committee on Engineering Acoustics, IEICE*, Mar. 2025. (in Japanese)
- [2] Daichi Kitamura, Tomohiko Nakamura, "Fundamentals and applications of audio source separation — A guide to becoming an expert," *2023 Otagaku Symposium*, Jun. 2023. (in Japanese)
- [3] Tomohiko Nakamura, "Signal-processing-inspired deep learning," *IEEE NZ Signal Processing/Information Theory Joint Chapter in co-hosted by the Acoustics Research Centre, University of Auckland*, Dec. 2022.
- [4] +1 invited presentation

Overview Papers.....

- [1] Shoichi Koyama, Juliano Ribeiro, Tomohiko Nakamura, Natsuki Ueno, and Mirco Pezzoli, "Physics-informed machine learning for sound field estimation: Fundamentals, state of the art, and challenges," *Special Issue on Model-Based and Data-Driven Audio Signal Processing, IEEE Signal Processing Magazine*, vol. 41, pp. 60–71, 2024.
- [2] Hirokazu Kameoka, Tomohiko Nakamura, and Norihiro Takamune, "Recent advances in music signal processing techniques," *The Journal of Institute of Electronics, Information and Communication Engineers*, vol. 98, no. 6, pp. 467–474, Jun. 2015. (in Japanese)

Patents.....

- [1] Tomohiko Nakamura, "Object recognition device, method, and program," Japan Patent JP7349288, 13-Sept-2023.
- [2] Tomohiko Nakamura, "Object recognition device, method, and program," Japan Patent JP7349290, 13-Sept-2023.
- [3] +9 patents

Awards

- 1. The Awaya Kiyoshi Research Award, ASJ, Mar. 2024.
- 2. The Itakura Prize Innovative Young Researcher Award, ASJ, Mar. 2022.
- 3. +12 awards